例题：

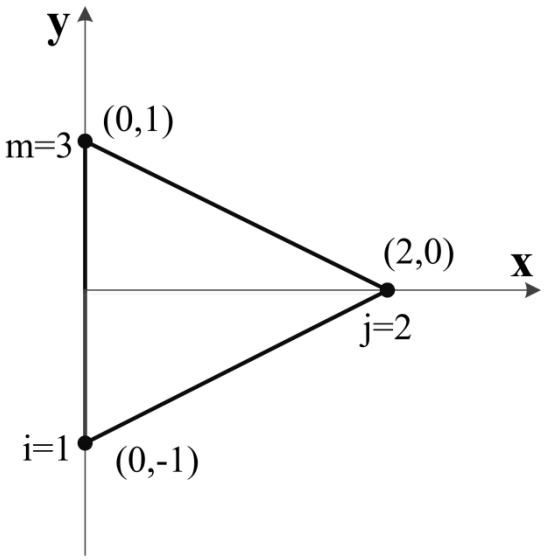
平面应力三角形单元如图所示。结点坐标单位为英寸（in ）。

设 $E=30\times10^{6}$ psi，$\mu=0.25$，厚度$t=1$ in。

1)计算单元的应变矩阵$[B]$和本构矩阵$[D]$

2)若给定单元的结点位移，求单元应力，主应力、主平面法矢量、主角度，Von Mises等效应力。

$u_1=0.0,\ v_1=0.0025$ in, $u_2=0.0012$ in, $v_2=0.0,\ u_3=0.0,\ v_3=0.0025$ in。



解：

对于平面应力三角形单元，由结点坐标可计算

$b_i=y_j-y_m=0-1=-1$

$b_j=y_m-y_i=1-(-1)=2$

$b_m=y_i-y_j=-1-0=-1$

$$c_i = -x_j + x_m = -2 + 0 = -2$$
$$c_j = -x_m + x_i = 0 + 0 = 0$$
$$c_m = -x_i + x_j = 0 + 2 = 2$$

$$a_i = x_j y_m - x_m y_j = 2\times 1 - 0\times 0 = 2$$
$$a_j = x_m y_i - x_i y_m = 0\times(-1) - 0\times 1 = 0$$
$$a_m = x_i y_j - x_j y_i = 0\times 0 - 2\times(-1) = 2$$

$$A = \frac{a_i + a_j + a_m}{2} = \frac{2+0+2}{2} = 2$$

应变矩阵:

$$[B] = \frac{1}{2A}\begin{bmatrix} b_i & 0 & b_j & 0 & b_m & 0 \\ 0 & c_i & 0 & c_j & 0 & c_m \\ c_i & b_i & c_j & b_j & c_m & b_m \end{bmatrix} = \frac{1}{2\times 2}\begin{bmatrix} -1 & 0 & 2 & 0 & -1 & 0 \\ 0 & -2 & 0 & 0 & 0 & 2 \\ -2 & -1 & 0 & 2 & 2 & -1 \end{bmatrix}$$

平面应力单元本构矩阵（弹性矩阵）:

$$[D] = \frac{E}{1-u^2}\begin{bmatrix} 1 & \mu & 0 \\ u & 1 & 0 \\ 0 & 0 & \dfrac{1-u}{2} \end{bmatrix} = \frac{30\times 10^6}{1-0.25^2}\begin{bmatrix} 1 & 0.25 & 0 \\ 0.25 & 1 & 0 \\ 0 & 0 & \dfrac{1-0.25}{2} \end{bmatrix} \text{ psi}$$

单元应力

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} = [D][B]\{\delta\}^e_{6\times 1}$$

$$= \frac{30\times 10^6}{1-0.25^2}\begin{bmatrix} 1 & 0.25 & 0 \\ 0.25 & 1 & 0 \\ 0 & 0 & 0.375 \end{bmatrix} \times \frac{1}{2\times 2}\begin{bmatrix} -1 & 0 & 2 & 0 & -1 & 0 \\ 0 & -2 & 0 & 0 & 0 & 2 \\ -2 & -1 & 0 & 2 & 2 & -1 \end{bmatrix} \begin{Bmatrix} 0.0 \\ 0.0025 \\ 0.0012 \\ 0.0 \\ 0.0 \\ 0.0025 \end{Bmatrix}$$

完成矩阵运算，可得

σ_x=19200 psi，σ_y=4800 psi，$\tau_{xy}=-15000$ psi

应力张量

$$\mathbf{S}=\begin{bmatrix}\sigma_x & \tau_{xy} & \tau_{xz}\\ \tau_{yx} & \sigma_y & \tau_{yz}\\ \tau_{zx} & \tau_{zy} & \sigma_z\end{bmatrix}=\begin{bmatrix}\sigma_x & \tau_{xy} & 0\\ \tau_{yx} & \sigma_y & 0\\ 0 & 0 & 0\end{bmatrix}=\begin{bmatrix}19200 & -15000 & 0\\ -15000 & 4800 & 0\\ 0 & 0 & 0\end{bmatrix}$$

解标准特征值问题

$\mathbf{Sn_p}=\sigma\mathbf{n_p}$

可得

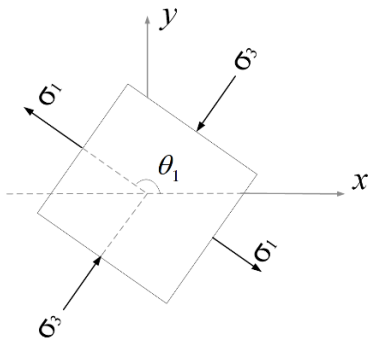
σ_1=28638.5;σ_2=0;$\sigma_3=-4638.5$;

$n_1=(-0.8464,0.5326,0)$;

$n_2=(0,0,1)$;

$n_3=(-0.5326,-0.8464,0)$;

$\theta_1=\arctan\left(\frac{0.5326}{-0.8464}\right)=147.8^\circ$ （称为主角度）



Von Mises等效应力

$$\sigma_e=\frac{1}{\sqrt{2}}\left[(\sigma_x-\sigma_y)^2+(\sigma_y-\sigma_z)^2+(\sigma_z-\sigma_x)^2+6(\tau_{xy}^2+\tau_{yz}^2+\tau_{zx}^2)\right]^{1/2}$$ =31217.6 psi

或

$$\sigma_e=\frac{1}{\sqrt{2}}\left[(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2\right]^{1/2}$$ =31217.6 psi